

List of Formulae
Slope of a line: $m = \frac{y_2 - y_1}{x_2 - x_1}$ Equation of a line: $y = mx + b$ OR $y - y_1 = m(x - x_1)$
Given $f(x) = ax^2 + bx + c$ is quadratic function, then: The equation of the axis of symmetry is: $x = -\frac{b}{2a}$ (x-coordinate of the vertex)
The <u>average rate of change</u> of a function $f(x)$ on the interval $[c, d]$ is: $\frac{f(d) - f(c)}{d - c}$
Solutions of a quadratic equation: $ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, a \neq 0$ Discriminant: $\Delta = b^2 - 4ac$